

Head-tail molecular clouds falling onto the Milky Way disk

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Abstract

We report discovery of two CO clouds which are likely falling down to the Galactic plane at more than 35 km s^{-1} . The clouds show head-tail distributions elongated perpendicular to the Galactic plane at $l = 331.6^\circ$ and $b = 0^\circ$ as revealed by an analysis of the Mopra CO $J=1-0$ survey data. We derived the distance of the clouds to be $2.46 \pm 0.18 \text{ kpc}$ based on the Gaia Data Release 3. The CO clouds have molecular masses of $4.8 \times 10^3 M_\odot$ and $3.5 \times 10^3 M_\odot$, respectively, and show kinetic temperature of 30–50 K as derived from the line intensities of the $^{13}\text{CO } J=2-1$, $^{12}\text{CO } J=1-0$, and $^{13}\text{CO } J=1-0$ emission. The temperature in the heads of the clouds is significantly higher than 10 K of the typical molecular clouds, although no radiative heat source is found inside or close to the clouds. Based on the results, we interpret that the present clouds are falling onto the Milky Way disk and are significantly heated up by the strong shock interaction with the disk HI gas. We suggest that the clouds represent part of the HI intermediate velocity clouds falling to the Galactic plane which were converted into molecular clouds by shock compression. This is the first case of falling CO clouds having direct observed signatures of the falling motion including clear directivity and shock heating. Possible implications of the CO clouds in the evolution of the Galactic interstellar medium are discussed.

Keywords: ISM: clouds — ISM: molecules — ISM: kinematics and dynamics — ISM: general — Galaxy: disk

1 Introduction

Vertical motions of the interstellar clouds relative to the Galactic plane can link the halo with the Galactic disk, whereas details of the vertical motions are not well understood because of difficulty in detecting motions perpendicular to the line of sight. A population of such clouds falling to the plane are the HI high velocity clouds (HVCs) having blue-shifted radial velocity from 40 km s^{-1} to 300 km s^{-1} . In particular, the low velocity HVCs in a velocity range of $40\text{--}100 \text{ km s}^{-1}$ are named the intermediate velocity clouds (IVCs), and one of them IVC 86-38 in the Pegasus–Pisces Arch (PP Arch) shows compelling evidence for falling clouds such as the head-tail distribution with a low dust-to-gas ratio (Fukui et

al. 2021), which is a unique case of the cloud-halo interaction as supported by numerical simulations of Shelton et al. (2022) (see also Kwak et al. 2009). Shelton et al. (2022) predicted that IVC 86-38 having HI mass of $\sim 10^4 M_\odot$ will be decelerated by the halo, and merge with the Galactic disk in $\sim 10 \text{ Myr}$, suggesting that most of the IVCs may be accreting onto the Galactic disk. While detections of CO lines in IVCs were reported previously (e.g., Desert et al. 1990; Weiß et al. 1999; Magnani & Smith 2010; Röhser et al. 2016), direct observational signatures of falling motion such as the moving direction and velocity were not revealed. We expect formation of H_2 and CO in the IVCs due to the high pressure near the plane. Studying such falling CO clouds on the Milky Way disk will be crucial in revealing the cloud properties and will shed light

on the evolution of the IVCs and star formation therein.

In the present paper, we report the results obtained by chance in the course of an observational investigation of the giant molecular cloud toward RCW 106 (Kohno et al. 2025). The RCW 106 (G333) GMC is located at $l = 330^\circ$ to $l = 335^\circ$ and exhibits active star formation. Toward the region we discovered small CO clouds of $\sim 10^3 M_\odot$ at velocity shifted by $\sim 15 \text{ km s}^{-1}$ from the GMC, and the purpose of the paper is to present the detailed observational properties of the clouds. The paper is organized as follows; Section 2 describes the datasets employed, Section 3 presents the results of the cloud distribution and their physical properties such as mass and temperature along with distance determination, and Section 4 discussion on the physical implications of the clouds along with the future direction of the research topic. Section 5 concludes the paper.

2 Data

2.1 Mopra Southern Galactic plane CO $J=1-0$ survey

We analyzed the CO $J=1-0$ data from the Mopra Southern Galactic Plane CO Survey (Burton et al. 2013; Braiding et al. 2015; Braiding et al. 2018; Cubuk et al. 2023). Mopra is a 22-m radio telescope operated by the Australia Telescope National Facility (ATNF) in Australia. It observed ^{12}CO , ^{13}CO , C^{18}O , and C^{17}O in the southern galactic plane during the winter seasons of 2011–2018. The observations were carried out in a rectangular unit of $60' \times 6'$ using the Fast-On-The-Fly (FOTF) mode. The front-end utilized a Monolithic Microwave Integrated Circuit (MMIC) receiver operating in the 3 mm band, and the back-end employed the UNSW Mopra Spectrometer (MOPS). A part of the survey observations from June 2018 to November 2018 were performed remotely from Nagoya in Japan through an internet connection by collaborators in the radio astronomy group at Nagoya University. The final spatial resolution at 115 GHz with the Mopra telescope is $33''$. The survey data were converted to main-beam temperature using the formula $T_{\text{MB}} = T_{\text{A}}^*/\eta$, with extended beam efficiency $\eta = 0.55$ (Ladd et al. 2005). The final spatial resolution of the cube data archived as Data Release 4 is $36''$, and the velocity resolution is 0.1 km s^{-1} . The cube data of Data release 4 of ^{12}CO and ^{13}CO $J=1-0$ used in the analysis were downloaded from the web page¹. The grid size of the data finally used was $(l, b, v) = (30'', 30'', 1 \text{ km s}^{-1})$, and the rms noise level was T_{mb} scale of $\sim 0.8 \text{ K}$ for ^{12}CO $J=1-0$ and $\sim 0.3 \text{ K}$ for ^{13}CO $J=1-0$, respectively.

2.2 SEDIGISM: APEX Galactic plane ^{13}CO $J=2-1$ survey

We also utilized archived data from the Atacama Pathfinder Experiment telescope (APEX), a 12m submillimeter telescope at an altitude of 5100 m in Chajnantor, Chile. The data were obtained by the SEDIGISM (Structure, Excitation and Dynamics of the Inner Galactic Interstellar Medium: Schuller et al. 2017, 2021; Duarte-Cabral et al. 2021) project. The receiver and spectrometer used in the observations were the Swedish Heterodyne Facility Instrument (SHFI) and the wide-band Fast Fourier Transform Spectrometer (XFFTS), respectively. The HPBW is $30''$ and the velocity resolution is 0.25 km s^{-1} . The data is calibrated to the T_{mb} scale, the grid size is $(l, b, v) = (9.5'', 9.5'', 0.25 \text{ km s}^{-1})$, and the r.m.s noise level is $\sim 0.8 \text{ K}$ at T_{mb} scale. We summarized the properties of CO line data using in this paper in Table 1.

¹ <https://mopracosurvey.wordpress.com>

2.3 The Herschel far-infrared data

We used the archival infrared image data obtained by Herschel² infrared Galactic Plane Survey (Hi-GAL: Molinari et al. 2010). The $160 \mu\text{m}$ image data is obtained by the Photodetector Array Camera and Spectrometer (PACS: Pilbratt et al. 2010). The spatial resolution of $160 \mu\text{m}$ image is $12''$.

2.4 Gaia Data Release 3

To estimate the distances to the head–tail molecular clouds, we used stellar data from Gaia Data Release 3 (Gaia Collaboration et al. 2023). We adopted “geometric distances” and G-band extinctions (A_G) from the catalog of Bailer-Jones et al. (2021). Uncertainties in A_G and geometric distance were taken as the 16th and 84th percentiles, representing the 68% confidence interval (Bailer-Jones et al. 2021; Gaia Collaboration et al. 2023).

3 Results

3.1 Spatial and velocity distributions of molecular clouds

In the course of the investigation of the GMC toward RCW 106 (Kohno et al. 2025), we discovered two head–tail molecular clouds elongated perpendicular to the Galactic plane at $l = 331.6^\circ$. The radial velocity of two clouds are separated from the GMC by $\sim 15 \text{ km s}^{-1}$. Figures 1(a) and 1(b) show the integrated intensity maps of the head–tail molecular clouds of ^{12}CO and ^{13}CO $J=1-0$ obtained by Mopra. Hereafter, we call these head–tail molecular clouds “Cloud 1” and “Cloud 2”, respectively. Cloud 1 has a peak at the Galactic plane $(l, b) = (331.65, -0.03)$ (the cloud head) and extends vertically over $\sim 0.4^\circ$ (the cloud tail). In the distribution of ^{13}CO , we can find that the peak position of Cloud 1 is $b \sim -0.05^\circ$, which is shifted to the negative b side of the Galactic plane. Cloud 2 has a peak at $(l, b) = (331.85, +0.20)$ (the cloud head) and has a C-shaped distribution extended toward the positive b side (the cloud tail).

We estimated the H_2 column density [$N(\text{H}_2)$] from the integrated intensity [$I_{12\text{CO}}$] of ^{12}CO $J=1-0$. $N(\text{H}_2)$ is given by equation (1),

$$N(\text{H}_2) = X_{\text{CO}} I_{12\text{CO}} [\text{cm}^{-2}], \quad (1)$$

where X_{CO} is the CO-to- H_2 conversion factor. In this paper, we used $X_{\text{CO}} = 2.0 \times 10^{20} [\text{cm}^{-2} (\text{K km s}^{-1})^{-1}]$ of the typical value in the Galactic disk (Bolatto et al. 2013; Sofue & Kohno 2020; Kohno & Sofue 2024). The peak column densities in the head of Cloud 1 and Cloud 2 are derived to be $N(\text{H}_2) \sim 4.0 \times 10^{21} \text{ cm}^{-2}$ and $N(\text{H}_2) \sim 4.2 \times 10^{21} \text{ cm}^{-2}$, respectively. In order to investigate the velocity distribution of Cloud 1 and Cloud 2, we made the intensity-weighted velocity (the first moment: V_c) and velocity dispersion (the second moment: σ_v) maps, by calculating the equations (2) and (3),

$$V_c = \frac{\int T_B(v) \cdot v dv}{\int T_B(v) dv} [\text{km s}^{-1}], \quad (2)$$

$$\sigma_v = \sqrt{\frac{\int T_B(v) \cdot (v - V_c)^2 dv}{\int T_B(v) dv}} [\text{km s}^{-1}], \quad (3)$$

where T_B is the brightness temperature and v is the radial velocity of the ^{12}CO $J=1-0$ emission. If a spectrum has a single

² Herschel Space Observatory is an ESA space observatory with science instruments provided by European-led Principal Investigator consortia and with important participation from NASA (Pilbratt et al. 2010)

Table 1. Properties of CO line data

Telescope	Line	HPBW	Grid	Velocity Resolution	r.m.s noise* level	References
Mopra	$^{12}\text{CO } J=1-0$	$\sim 33''$	$30''$	0.1 km s^{-1}	$\sim 0.8 \text{ K}$	[1,2,3,4]
	$^{13}\text{CO } J=1-0$	$\sim 33''$	$30''$	0.1 km s^{-1}	$\sim 0.3 \text{ K}$	[1,2,3,4]
APEX	$^{13}\text{CO } J=2-1$	$\sim 30''$	$9''.5$	0.25 km s^{-1}	$\sim 0.8 \text{ K}$	[5,6,7]

Note. *The r.m.s noise level is taken from the final cube data using this paper.

References [1] Burton et al. (2013), [2] Braiding et al. (2015), [3] Braiding et al. (2018), [4] Cubuk et al. (2023), [5] Schuller et al. (2017), [6] Schuller et al. (2021), [7] Duarte-Cabral et al. (2021)

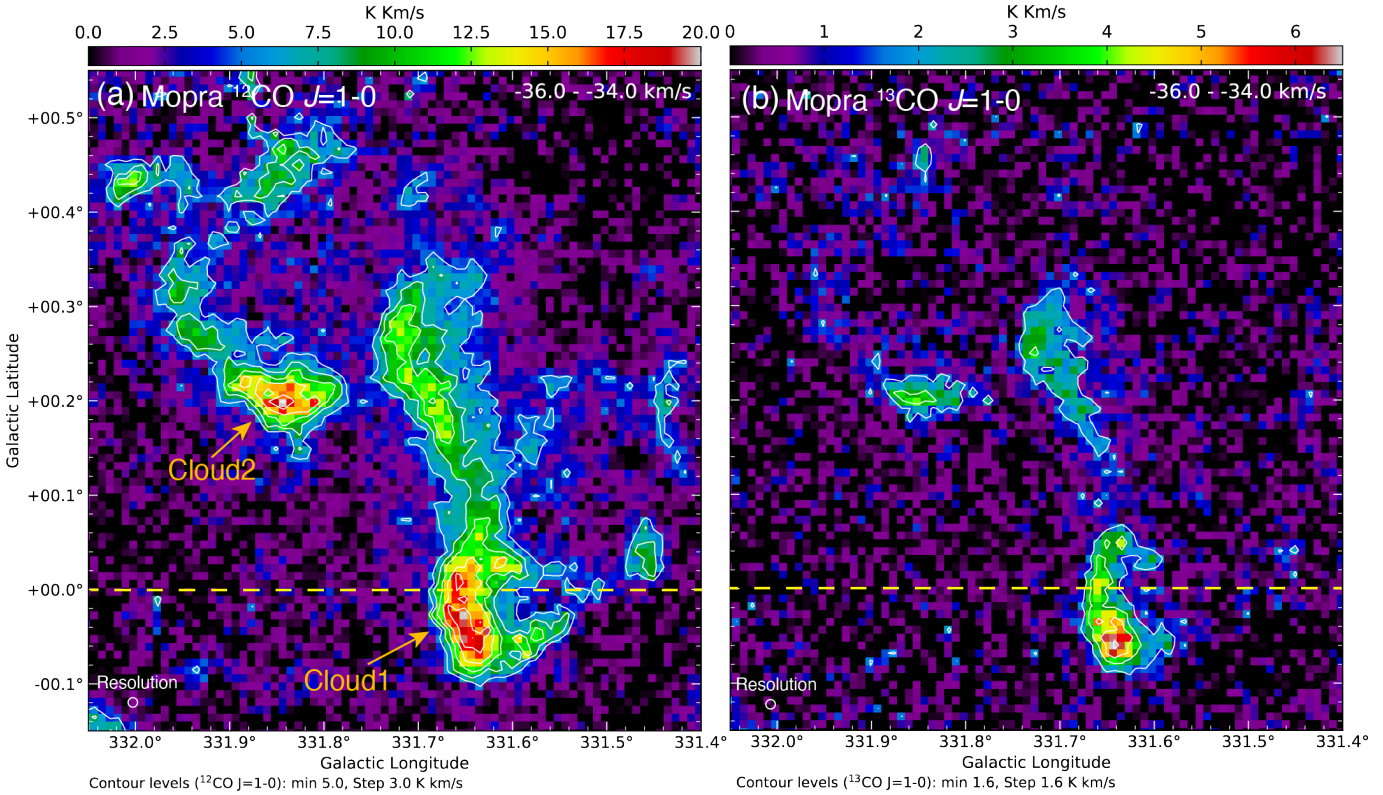


Fig. 1. (a) The Mopra $^{12}\text{CO } J=1-0$ integrated intensity map of the head-tailed molecular clouds. The lowest contour levels and intervals are 5.0 K km s^{-1} and 3.0 K km s^{-1} . (b) Same as (a), but for $^{13}\text{CO } J=1-0$. The lowest contour levels and intervals are 1.6 K km s^{-1} and 1.6 K km s^{-1} . The integrated velocity ranges are from -36 km s^{-1} to -34 km s^{-1} . The yellow dotted lines show the Galactic plane ($b = 0^\circ$). Alt text: Mopra 12-CO and 13-CO integrated intensity maps.

Gaussian shape, the relation between σ_v and full-width half maximum (FWHM) line-width ΔV is $\Delta V = \sqrt{8 \ln 2} \sigma_v$. Figures 2(a) and 2(b) present the first and the second moment maps, respectively. These moment maps were created for a velocity range from -40 km s^{-1} to -30 km s^{-1} . In Figure 2 (a), Cloud 1 is distributed with a uniform radial velocity of $\sim -35 \text{ km s}^{-1}$. Cloud 2 has a velocity range from -39 km s^{-1} to -36 km s^{-1} . We can find that the velocity dispersion of head in Cloud 1 has $1.4\text{--}1.5 \text{ km s}^{-1}$, which is larger than the tail having $\sim 0.8 \text{ km s}^{-1}$ as shown in Figure 2(b). In particular, Cloud 2 has a large velocity dispersion of $2.0\text{--}2.5 \text{ km s}^{-1}$ at the intensity peak in the head part and a velocity dispersion of $0.3\text{--}0.7 \text{ km s}^{-1}$ in the tail part.

3.2 The $^{13}\text{CO } J=2-1/1-0$ intensity ratio

We analyzed the intensity ratio between $^{13}\text{CO } J=2-1$ and $^{13}\text{CO } J=1-0$ (hereafter $R_{2-1/1-0}^{13}$) in order to derive the physical condi-

tion of the molecular clouds. The CO line intensity ratio between different rotational excitation levels reflects the kinetic temperature and density of the molecular gas. The spatial resolution of the $^{13}\text{CO } J=2-1$ and $^{13}\text{CO } J=1-0$ data are smoothed to be $120''$ to improve signal-to-noise ratios. The grid size of $^{13}\text{CO } J=2-1$ data obtained by APEX is aligned to $^{13}\text{CO } J=1-0$ data taken from Mopra by using the miriad software. Figure 3 shows the $R_{2-1/1-0}^{13}$ map. The intensity ratio map was created above 1.7 K km s^{-1} in the $^{13}\text{CO } J=2-1$ and $^{13}\text{CO } J=1-0$ integrated intensities. $R_{2-1/1-0}^{13}$ is enhanced in the head and tail of Cloud 1 to be $R_{2-1/1-0}^{13} \sim 0.9$ and $R_{2-1/1-0}^{13} \sim 0.85$, while it has low of $R_{2-1/1-0}^{13} \sim 0.5$ around $(l, b) = (331.67^\circ, 0.02^\circ)$. In the head of Cloud 2, the ratio is enhanced to be $R_{2-1/1-0}^{13} \sim 1.1$.

We calculated the kinetic temperature [T_{kin}] and molecular hydrogen number density [$n(\text{H}_2)$] by assuming the Large Velocity Gradient (LVG) model (Goldreich & Kwan 1974; Scoville & Solomon 1974). The LVG calculation assumes a spherical molec-

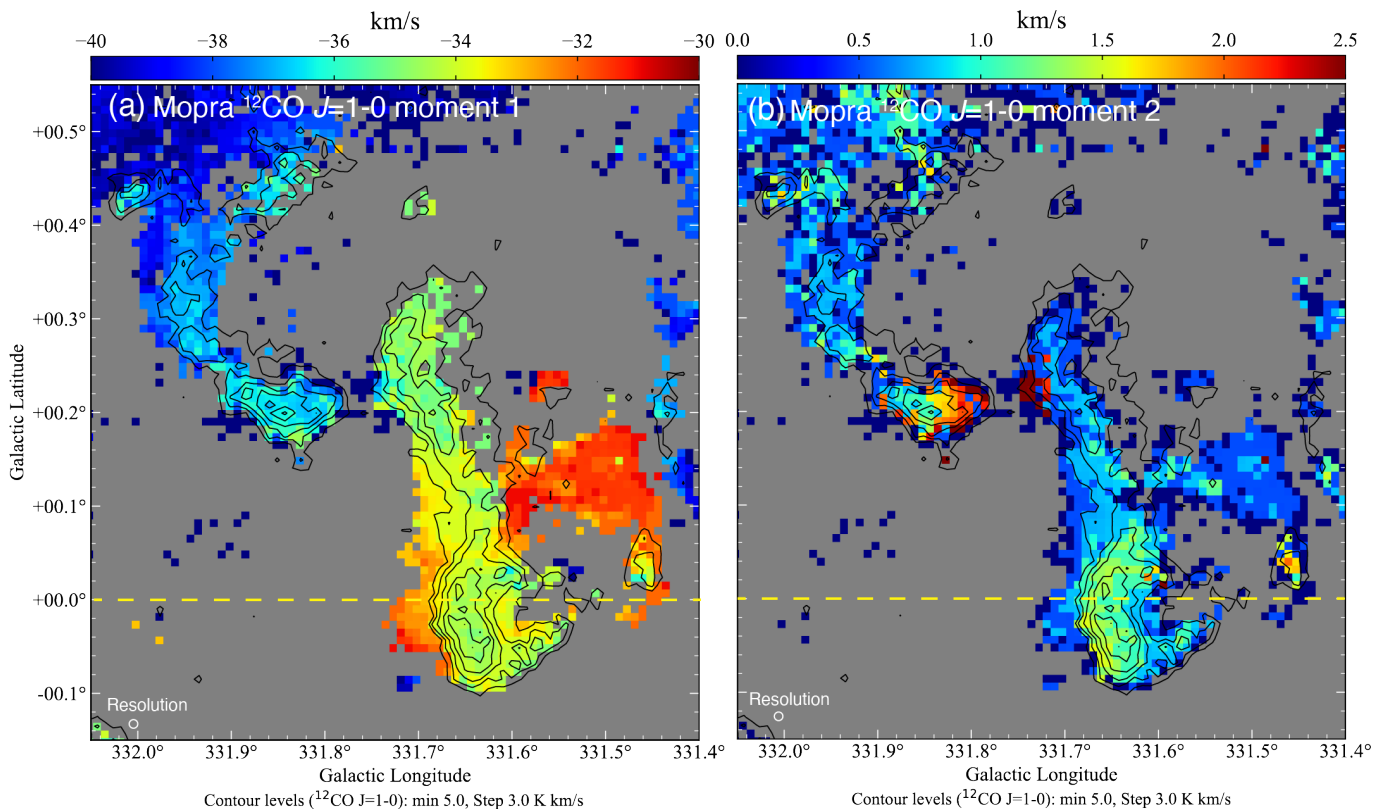


Fig. 2. The Mopra $^{12}\text{CO } J = 1-0$ velocity-field (first-moment) map. (b) The $^{12}\text{CO } J = 1-0$ velocity dispersion (second-moment) map. The adopted velocity range extends from -40 km s^{-1} to -30 km s^{-1} . The yellow dotted lines show the Galactic plane ($b = 0^\circ$). The contour levels and intervals are the same as in Figure 1(a). The data points are plotted above 4.5 K km s^{-1} of the $^{12}\text{CO } J = 1-0$ integrated intensity. Alt text: Mopra 12-CO first and second moment maps.

Table 2. Results of the LVG analysis of the head-tail molecular clouds.

Name	l [deg]	b [deg]	ΔV [km s $^{-1}$]	r [pc]	$R_{2-1/1-0}^{13}$	$R_{1-0}^{13/12}$	$n(\text{H}_2)$ [cm $^{-3}$]	T_{kin} [K]
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
A	331.719	+0.240	2.2	2.7	1.19	0.23	$(2.0^{+1.0}_{-0.7}) \times 10^3$	47^{+21}_{-18}
B	331.660	+0.073	2.1	2.1	0.77	0.27	$(1.1^{+0.4}_{-0.3}) \times 10^3$	25^{+13}_{-9}
C	331.619	-0.069	2.8	1.8	0.92	0.34	$(1.5^{+0.6}_{-0.4}) \times 10^3$	29^{+13}_{-10}
D	331.844	+0.190	3.8	1.7	1.22	0.15	$(2.0^{+1.3}_{-0.9}) \times 10^3$	41^{+26}_{-19}

Note. (1) Position name (2) Galactic longitude (3) Galactic latitude (4) FWHM of the $^{13}\text{CO } J = 1-0$ spectra. (5) Radius defined by $\sqrt{S/\pi}$ above the 30% level of the peak $^{13}\text{CO } J = 1-0$ integrated intensity. The boundary between B and C is decided at $b = 0$. (6) Intensity ratio between $^{13}\text{CO } J = 2-1$ and $^{13}\text{CO } J = 1-0$. (7) Intensity ratio between $^{13}\text{CO } J = 1-0$ and $^{12}\text{CO } J = 1-0$. (8) The number densities of molecular hydrogen. (9) The kinetic temperatures.

ular cloud with a constant velocity gradient (dv/dr), and uniform density and temperature. It solves the statistical equilibrium equations at each rotational energy level by considering the effect of optical depth with the photon escape probability (Castor 1970).

Figure 4 shows the spectra and results of the LVG calculations for data points of A, B, C, and D as shown in Figure 3. The calculations used the integrated intensity ratio in the velocity range shown by the vertical dotted lines of the spectrum. We assumed that the CO and H_2 abundance ratio is given as $[^{12}\text{CO}]/[\text{H}_2] = 10^{-4}$ (Frerking et al. 1982; Pineda et al. 2010) and the isotope abundance ratio is $[^{12}\text{C}]/[^{13}\text{C}] = 53$ (4 kpc ring in Table 4 of Wilson & Rood 1994). The velocity gradient (dv/dr) was calculated from

the FWHM and cloud radius (r). FWHM is estimated by a single Gaussian fitting to each $^{13}\text{CO } J = 1-0$ spectrum. r is derived by $\sqrt{S/\pi}$, where S is the cloud area above the 30% level of the peak integrated intensity. By the analysis we derived that Cloud 1 and Cloud 2 have $T_{\text{kin}} = 25-50 \text{ K}$ and $n(\text{H}_2) = (1.0-2.0) \times 10^3 \text{ cm}^{-3}$. In particular, the head and tail parts (A, B, and D) have high temperatures as compared with 10 K, the typical kinetic temperature of molecular gas without extra heating sources (e.g., Nishimura et al. 2015). The results of the LVG calculations at each point A, B, C, and D are summarized in Table 2.

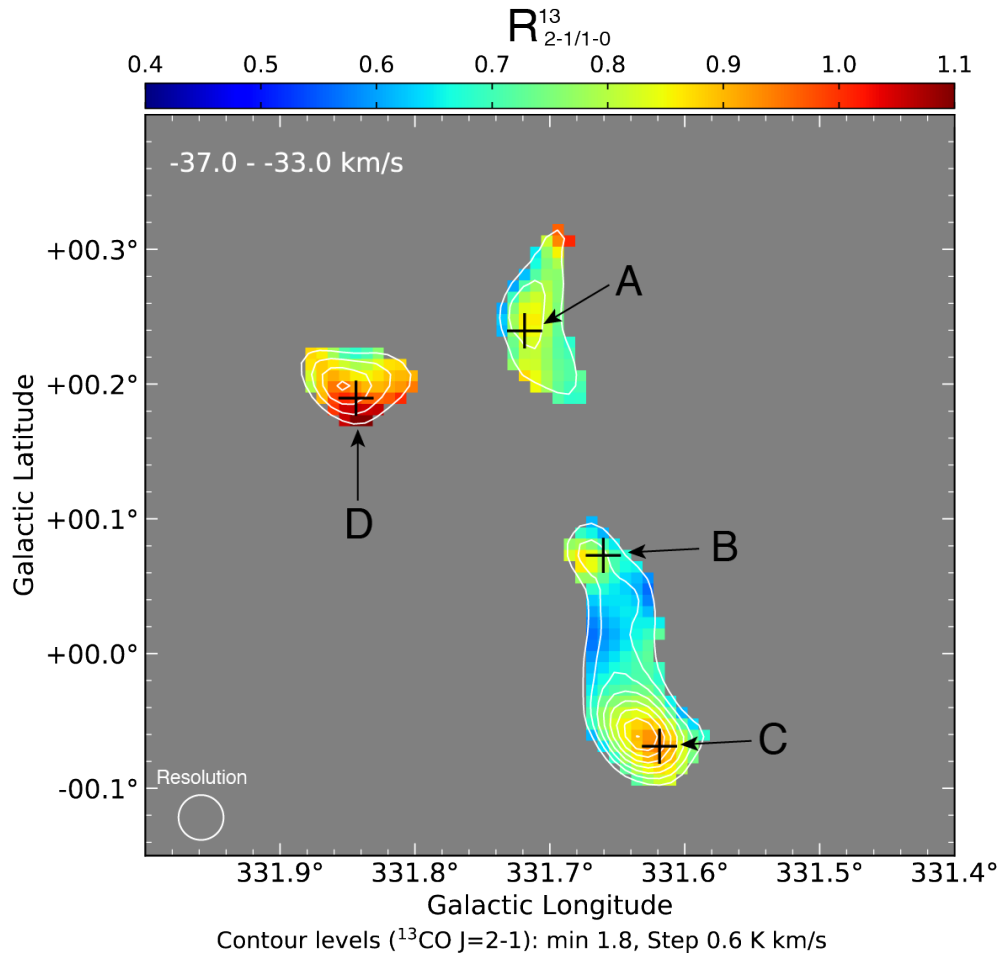


Fig. 3. The $^{13}\text{CO } J = 2-1/1-0$ ratio map above the integrated intensity of 1.8 K km s^{-1} . The integrated velocity range is from -37 km s^{-1} to -32 km s^{-1} . The data are smoothed to the spatial resolution of $120''$. The lowest contour level and interval are 1.8 K km s^{-1} and 0.6 K km s^{-1} of $^{13}\text{CO } J = 2-1$, respectively. A, B, C, and D indicate the position of the LVG calculation as shown in Figure 4 and Table 2. Alt text: The ^{13}CO intensity ratio map.

3.3 Comparison with the infrared image

We compared the CO data with Herschel infrared images to investigate the origin of the heating in the head-tail molecular clouds. Figure 5(a) shows the spatial distribution of the head-tail molecular clouds overlaid on the Herschel $160 \mu\text{m}$ continuum image (Molinari et al. 2010). Figure 5(b) presents a close-up map image of the head part in Cloud 1. The infrared peak close to the tip of Cloud 1 is the massive star-forming region G331.5-0.1 associated with the Norma arm at a distance of 7.5 kpc (Merello et al. 2013), which exists at the background source toward the head-tail molecular clouds as discussed in Section 4.1. So, we find no infrared source toward Cloud 1 and Cloud 2 which have high kinetic temperatures, $T_{\text{kin}} = 47^{+21}_{-18} \text{ K}$ of and $T_{\text{kin}} = 41^{+26}_{-19} \text{ K}$, respectively. This indicates that there is no responsible radiative heating source for the high temperature in the region of Cloud 1 and Cloud 2.

4 Discussion

4.1 Distance to the head-tail molecular clouds

In order to investigate the properties of the present clouds, we need a reliable distance estimate. Figure 6(a) shows a face-on schematic view of the Milky Way, and the red arrow indicates the line of sight at $l = 331^\circ$. The Norma arm and Scutum-Centaurus arm overlap in

this direction. Figure 6(b) shows the spectrum at the head of Cloud 1. The velocity component from -100 km s^{-1} to -80 km s^{-1} corresponds to the Norma arm, which is associated with the massive star-forming region G331.5-0.1 (Merello et al. 2013). The velocity component from -80 km s^{-1} to -50 km s^{-1} is associated with the RCW 106 (G333) GMC complex at the Scutum-Centaurus arm (Nguyen et al. 2015; Zhou et al. 2023; Kohno et al. 2025). The head-tail molecular clouds have -35 km s^{-1} , which is red-shifted by more than 15 km s^{-1} from these spiral arms and do not correspond to the main velocity component in the spiral arms in the direction of $l = 331^\circ$ (Vallée 2017).

In order to estimate the distance to the head-tail clouds, we analyzed the data of fixed stars obtained by Gaia Data Release 3 (Gaia Collaboration et al. 2023). In this paper, we used a stellar distance catalog based on Bailer-Jones et al. (2021) to attempt to estimate the distance of the head-tail molecular clouds from the G-band extinction (A_G) and annual parallax measurements. Figure 7 presents scatter plots between A_G and the distance of Gaia sources corresponding to Cloud 1, which has the $^{12}\text{CO } J = 1-0$ integrated intensity greater than 5 K km s^{-1} . If the interstellar dust is distributed uniformly, A_G increases with distance. However there is a dense molecular cloud, A_G has a breaking point. Thus, we can estimate the line-of-sight distance of the molecular cloud, by analyzing the breaking points of A_G (e.g., Yan et al. 2019; Zucker et

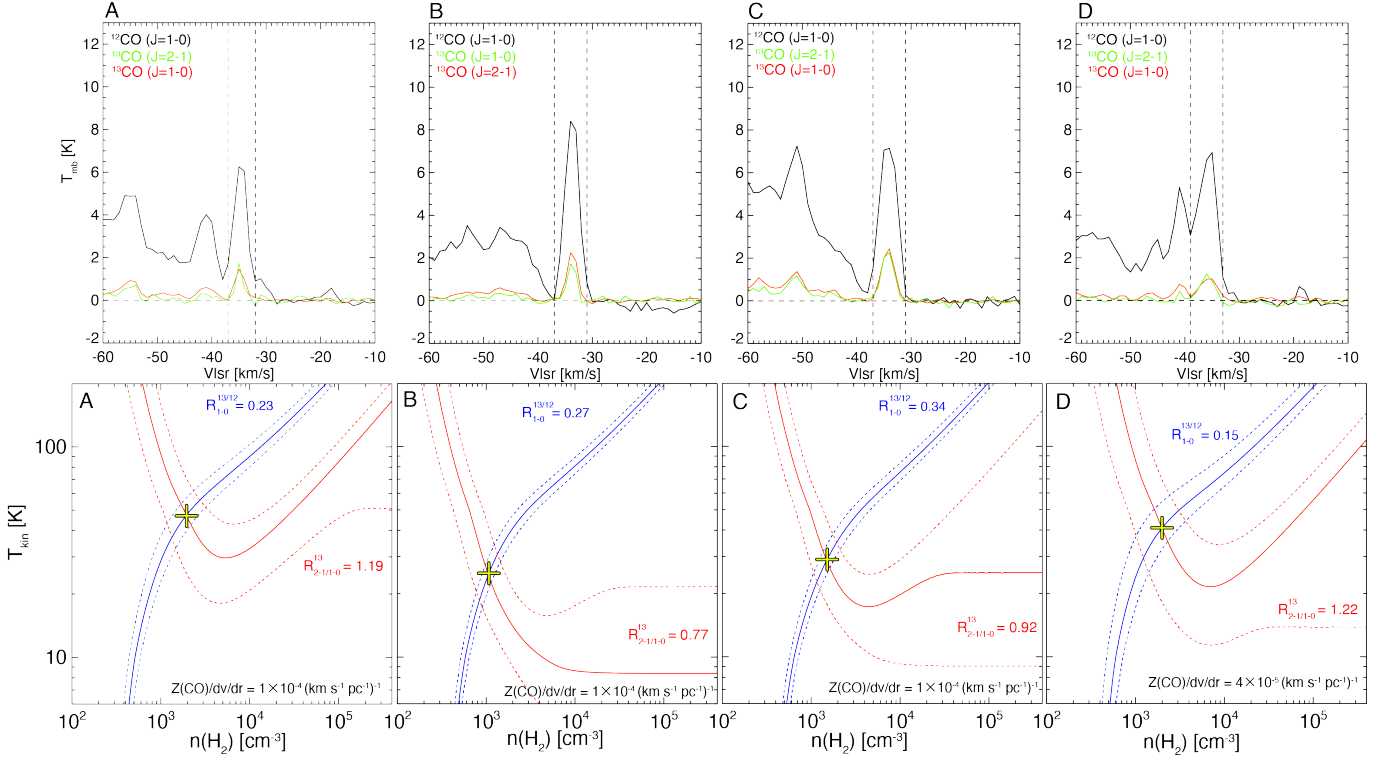


Fig. 4. Spectra and results of the LVG calculation at the point A, B, C, and D. The black, green, and red spectra show the $^{12}\text{CO } J=1-0$, $^{13}\text{CO } J=2-1$, and $^{13}\text{CO } J=1-0$, respectively. The vertical dotted lines indicate the integrated velocity range. Blue and red curves of constant $R_{13/12}$ and $R_{13/2-1}$ as functions of the molecular gas density [$n(\text{H}_2)$] and kinetic temperature [T_{kin}]. $Z(\text{CO})$ is the CO and H_2 abundance ratio given by $[^{12}\text{CO}]/[\text{H}_2] = 10^{-4}$ (Frerking et al. 1982; Pineda et al. 2010). The dotted lines show the $\pm 1 \sigma$ error of the intensity ratio. The yellow cross marks indicate the solution of each data point. Alt text: Spectra and results of LVG calculation at the point A, B, C, and D.

al. 2019; Doi et al. 2021; Doi et al. 2024). Break point analysis methods that apply to Gaia DR3 in this paper were presented by Doi et al. (2021) and in Appendix C of Doi et al. (2024). Based on the analysis of the break points, we found that there are three break points at distances of 1.11 ± 0.09 kpc, 1.44 ± 0.09 kpc, and 2.46 ± 0.18 kpc, as shown by the vertical red dotted lines in Figure 7. In order to determine the distance, we investigated the spatial distribution of Gaia sources at these three breaking points.

Figures 8 (a), (b), and (c) show stellar density maps of Gaia sources at distances from 1.02 kpc to 1.35 kpc, from 1.35 kpc to 2.28 kpc, and from 2.28 kpc to 4.0 kpc, respectively. The right panels of each figure show $^{12}\text{CO } J=1-0$ contours overlaid on the stellar density map. The boundaries at 1.02 kpc, 1.35 kpc, and 2.28 kpc correspond to the distance of the lower limit at the three breaking points. We cannot find density distributions corresponding to Cloud 1 in the panels at distances from 1.02 kpc to 1.35 kpc and from 1.35 kpc to 2.28 kpc as shown in Figure 8 (a) and (b). On the other hand, in the panel from 2.28 kpc to 4.0 kpc, the extinction of stellar density has morphological correspondence with CO distributions at the tips of Cloud 1 (Figure 8c). These results indicate that Cloud 1 exists at a distance within a range from 2.28 kpc to 4 kpc. Thus, we suggest that 2.46 ± 0.18 kpc is the most likely distance to Cloud 1 among the three breaking points. If we adopt distance $d = 2.46$ kpc, the length of Cloud 1 in $^{12}\text{CO } J=1-0$ is ~ 15 pc and its width is ~ 5 pc. The total molecular mass (M_{cloud}) is given by

$$M_{\text{cloud}} = \mu_{\text{H}_2} m_{\text{H}} d^2 \Omega \sum_i N_i(\text{H}_2) [\text{M}_{\odot}], \quad (4)$$

where Ω is the solid angle subtended by the molecular cloud, μ_{H_2} the average molecular weight of a hydrogen molecule, 2.8, m_{H} the proton mass, 1.67×10^{-24} g, and $N_i(\text{H}_2)$ the hydrogen molecular column density at the i -th pixel. The molecular cloud mass of Cloud 1 is calculated as $4.8 \times 10^3 M_{\odot}$. The CO distributions of Cloud 2 also seem to fit the stellar density map from 1.35 kpc to 2.28 kpc as well as from 2.28 kpc to 4.0 kpc. On the other hand, the radial velocities of Cloud 1 and Cloud 2 are in a common velocity range from -36 km s^{-1} to -34 km s^{-1} . This result shows that Cloud 2 is likely to be at the same distance 2.46 ± 0.18 kpc estimated for Cloud 1, while we cannot exclude that Cloud 2 possibly exists with a foreground distance of 1.44 ± 0.09 kpc. If we assume 2.46 kpc to the distance of Cloud 2, its length, width, and total mass are derived to be ~ 10 pc, ~ 3 pc, and $3.5 \times 10^3 M_{\odot}$. We note that the difference in the possible distance of Cloud 2 does not affect the falling cloud scenario onto the Milky Way disk that we discuss in the following sections.

4.2 Interaction of the head-tail molecular clouds with the gas disk component

We present position-velocity diagrams toward Cloud 1 and Cloud 2 in order to investigate the relationship with the gas disk components. Figure 9(a) shows the spatial distribution of the head-tail molecular clouds. Figures 9(b) and (c) present latitude-velocity diagrams of Cloud 1 and Cloud 2, respectively. Kohno et al. (2025) reported the three velocity components at -64 km s^{-1} , -54 km s^{-1} , and -44 km s^{-1} toward RCW 106 West, which includes

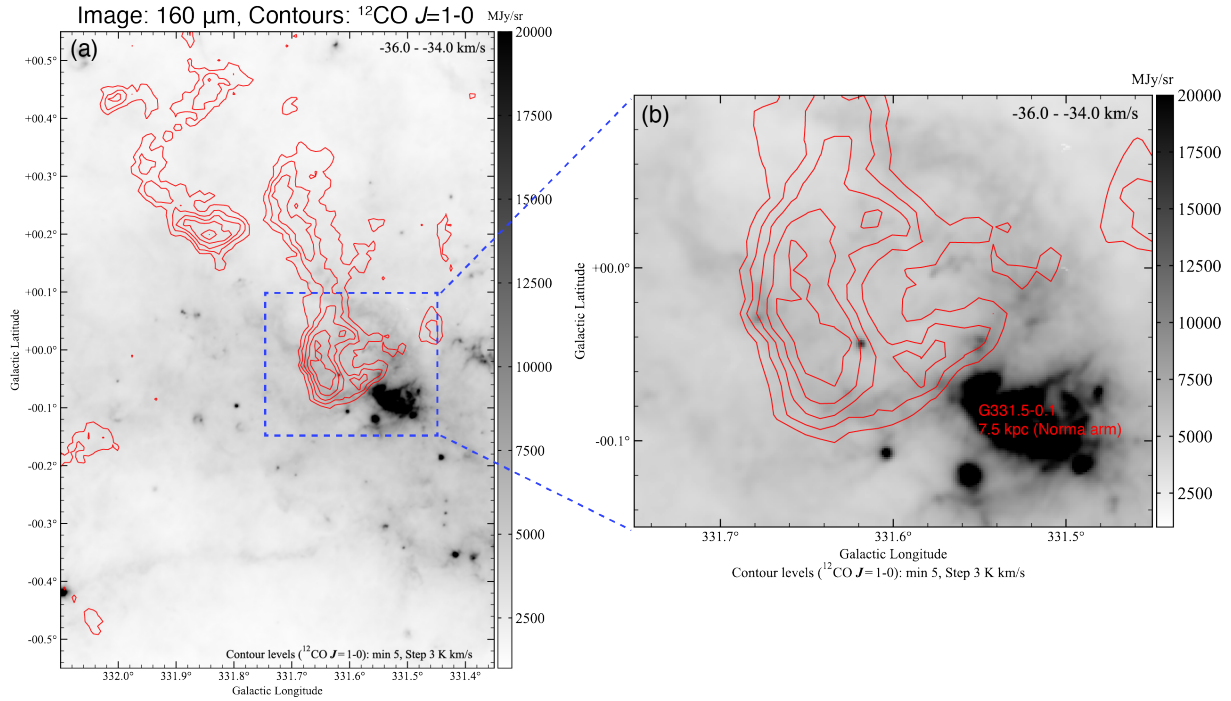


Fig. 5. (a) $^{12}\text{CO } J=1-0$ spatial distributions of head-tail molecular clouds superposed on the Herschel $160\ \mu\text{m}$ continuum image (Molinari et al. 2010). The lowest contour levels and intervals are $5.0\ \text{K km s}^{-1}$ and $3.0\ \text{K km s}^{-1}$. (b) The close-up image of panel (a). Alt text: Mopra ^{12}CO spatial distributions of head-tail molecular clouds superposed on the Herschel $160\ \mu\text{m}$ continuum image.

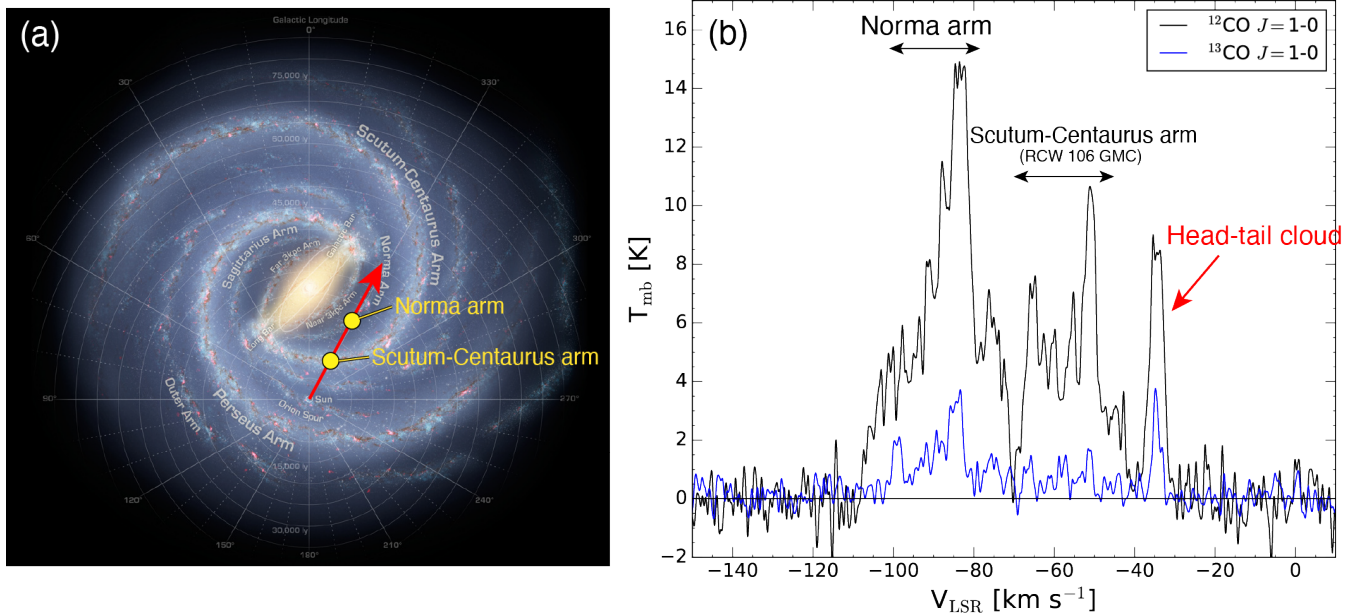


Fig. 6. (a) The top schematic view of the Milky Way (NASA/JPL-Caltech/ESO/R. Hurt). The red arrow indicates the direction of $l = 331^\circ$. Two yellow circles are present at the intersections of the line-of-sight with the Scutum-Centaurus and Norma arms. (b) Spectra of Mopra ^{12}CO and $^{13}\text{CO } J=1-0$ at $(l, b) = (331.^\circ 627, -0.^\circ 0604)$. Alt text: The top schematic view of the Milky Way and spectra of Mopra ^{12}CO and ^{13}CO .

the region analyzed in the present paper (see Figures 9 and 10 in Kohno et al. 2025). The authors suggested that the $-44\ \text{km s}^{-1}$ component with a velocity range from $-50\ \text{km s}^{-1}$ to $-38\ \text{km s}^{-1}$ is a diffuse foreground component different from the RCW 106 GMC complex at a distance of 3.6 kpc, and exists at the front face of the Scutum-Centaurus arm. A previous study of Russeil et

al. (2005) also supports the interpretation of the diffuse component based on a velocity-resolved deep $\text{H}\alpha$ survey. Cloud 1 component connects to the $-44\ \text{km s}^{-1}$ component with the bridge features as shown by red arrows in Figure 9(b). We also find Cloud 2 as a compact peak in Figure 9(c), which merges with the $-44\ \text{km s}^{-1}$ component. The numerical simulations of cloud-cloud colli-

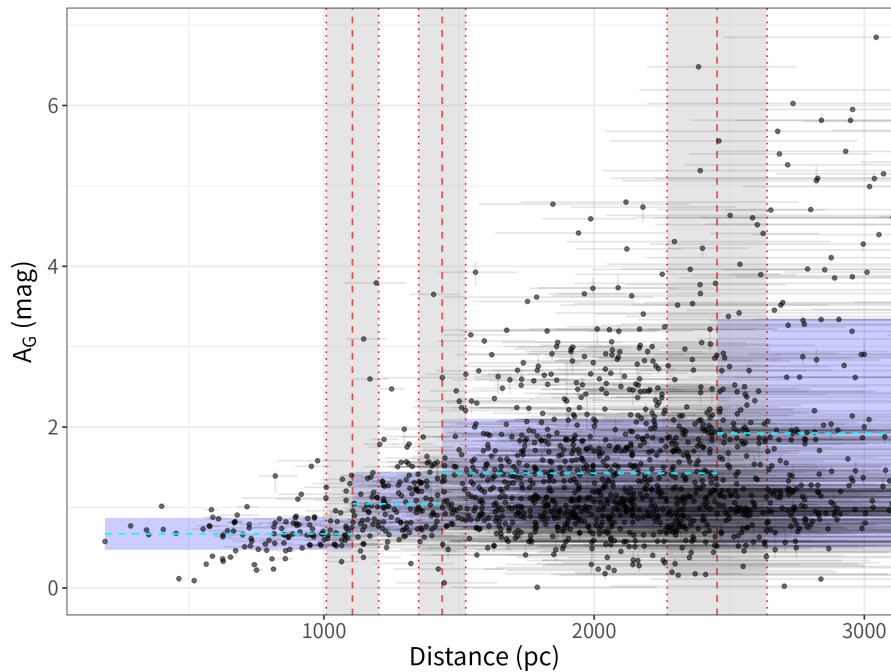


Fig. 7. The scatter plot between A_G and the distance of fixed stars corresponding to Cloud 1. The vertical red bold lines indicate the breakpoints of A_G . The vertical grey shadow areas show the ranges of $\pm 1\sigma$ standard deviation in each break point. The horizontal light blue dotted line and purple shadow area show the median and mean absolute deviation ranges of A_G divided into three break points, respectively. The mean absolute deviation values are scaled to 1σ standard deviation. The uncertainties of A_G are their 16th and 84th percentiles from the Gaia DR3 catalog (Gaia Collaboration et al. 2023). The errors of the geometric distance are their 16th and 84th percentiles, which correspond to a 68% confidence interval (Bailer-Jones et al. 2021). Alt text: The scatter plot between G-band extinction and the distance of fixed stars corresponding to Cloud 1 obtained by Gaia DR3.

sions (Takahira et al. 2014, 2018) reproduce bridge features and merging two velocity components in a position-velocity diagram of the collision between two molecular clouds (see also Figure 9 in Haworth et al. 2015 and Figure 10 in Torii et al. 2017). Based on these results, we propose that head-tail molecular clouds interact with the Galactic gas disk component at the front face of the Scutum–Centaurus arm.

4.3 The falling scenario onto the Milky Way disk

We first discuss the morphological properties of the clouds. We suggest that the morphology of the head-tail molecular clouds is explained by falling motion onto the Galactic disk. Indeed, the magnetohydrodynamic (MHD) simulations were reproduced with the head-tail morphology toward the Galactic plane by falling motion onto the disk from the Galactic halo (see Figure 7 in Kwak et al. 2009, Shelton et al. 2022). The former simulation shows that the falling cloud has a U-shaped shocked region with enhanced density and turbulence in the tip of the head. These are similar properties to the present clouds, and we propose that the origin of the heating and large velocity dispersion at the head of head-tail molecular clouds were caused by the shock compression and heating by the interactions with the Galactic disk. We point out that the head-tail cloud in their simulation by Kwak et al. (2009) having a size of ~ 3 kpc from the head to the tail. This size is a factor ~ 200 larger than the head-tail molecular cloud reported in this paper, but their results are useful to infer the correspondence of morphologies between observation and simulation since the local shock interactions are not dominated by gravity and are scale-free. More realistic simulations of smaller molecular clouds in the fu-

ture will be useful in understanding more detailed correspondence in temperature, etc.

The origin of the falling clouds needs to be explored. As discussed in the introduction, in the 1960’s HI observations revealed the high-velocity clouds (HVCs) and the intermediate velocity clouds (IVCs) at high Galactic latitudes have radial velocity of $|V_{\text{LSR}}| \gtrsim 50\text{--}100$ km s $^{-1}$ without rotation (Muller et al. 1963; Smith 1963; Oort 1970; Wakker & van Woerden 1997; Bland-Hawthorn et al. 1998; Blitz et al. 1999; Wakker 2001; Sofue et al. 2004; Fraternali et al. 2015; Lockman et al. 2008; Liu et al. 2025). It is possible that the present falling clouds are linked with the IVCs because their velocities are in a similar range. The origins of the IVCs are thought to be the accretion of gas ejected from the Galactic disk into the halo by the feedback of supernova explosions called Galactic Fountains (Bregman 1980; Marasco et al. 2013), or low-metallicity atomic gas falling into the Galactic disk from outside the Galaxy (Fukui et al. 2021; Hayakawa & Fukui 2024). In particular, it is worth noting that the absorption line measurements supporting the Galactic Fountains were made in only several lines of sight toward the bright background stars/galaxies in the IVCs. This makes a strong contrast with the observations of dust-to-gas ratio made by Hayakawa & Fukui (2024), which covers consecutively the whole IVCs at b higher than 15 degrees. We argue that the low metallicity IVCs from the outside the Milky Way are falling on to the Milky Way disk and might be forming to molecular gas like head-tail molecular clouds presented in this paper.

The recent study of an IVC named IVC 86-36 in PP Arch has a head-tail morphology in the radial velocity from -50 to -30 km s $^{-1}$, and the total cloud mass of $\sim 7 \times 10^3 M_{\odot}$ (Fukui et al. 2021). The mass is comparable to those of Cloud 1 and Cloud

2, while the size of the head of IVC 86-36 is ~ 90 pc, an order of magnitude larger than the present clouds. Shelton et al. (2022) showed that the observational features of IVC 86-36 can be explained by collision and merging with the Galactic halo-disk from their numerical simulations. We therefore propose that the head-tail molecular clouds are falling onto the Galactic disk in a similar way to the IVC 86-36, while a process which converts the HI gas into molecular form is required to explain the CO clouds. It is possible that the conversion of HI into H_2 occurs at a lower height of $\ll 1$ kpc from the plane, where the ambient pressure due to the HI gas disk can become high enough to compress HI to high density more than 10^3 cm^{-3} . At the density, HI can be converted to H_2 via dust surface reaction in Myr (e.g., Hollenbach & Salpeter 1971). It is also to be noted that the estimated temperature of the clouds is as high as 30-50 K. This suggests that the shock heating is fairly strong, involving high-speed collision at a relative velocity higher than 35 km s^{-1} , which is consistent with the IVC velocity.

A few other examples of a head-tail cloud with a head-tail distribution were discovered in the N159 region of the Large Magellanic Cloud (LMC), where the clouds likely originated from high-velocity HI gas colliding onto the LMC disk at $> 50 \text{ km s}^{-1}$ (Fukui et al. 2019; Tokuda et al. 2019, 2022). These clouds were interpreted based on the MHD simulations of a falling cloud by Inoue et al. (2018) (see also for visualization of the results Maity et al. 2024). The clouds show heated head regions, while the heating was made radiatively by the high mass stars formed instead of the shock heating. The observed shock velocity in N159 is $\sim 20 \text{ km s}^{-1}$, probably somewhat lower than the case of the IVC origin, and the shock heating does not dominate the heating.

The discovery of the falling CO clouds has revealed for the first time that dense CO clouds are falling down to the Galaxy at high velocity. While detections of CO lines in IVCs were reported previously (e.g., Desert et al. 1990; Weiß et al. 1999; Magnani & Smith 2010; Röhser et al. 2016), direct observational signatures of falling motion such as the moving direction and velocity were not revealed. The discovery offers a tight connection between the Galactic halo and disk, and can connect the Milky Way and the inter-galactic matter. The discovery raised the following issues to be addressed in future studies;

- Are the falling CO clouds a common phenomenon in the Galaxy? If yes, what is the fraction of the total mass included in the falling CO clouds.
- Whether the origin of the falling clouds is the HI gas in the IVCs, and whether the falling motion follows that of the IVCs.
- Are the falling CO clouds forming stars? What are the impacts of the falling clouds on star formation?
- What is the role of the falling clouds in cycling the ISM in the Galaxy?

Observational and theoretical studies on these issues will be beneficial to better understand the galaxy evolution in a three-dimensional perspective, while the conventional perspective is mainly limited to a thin layer of the Galactic plane. The three-dimensional view may suggest the Galaxy as a system more open to the intergalactic space.

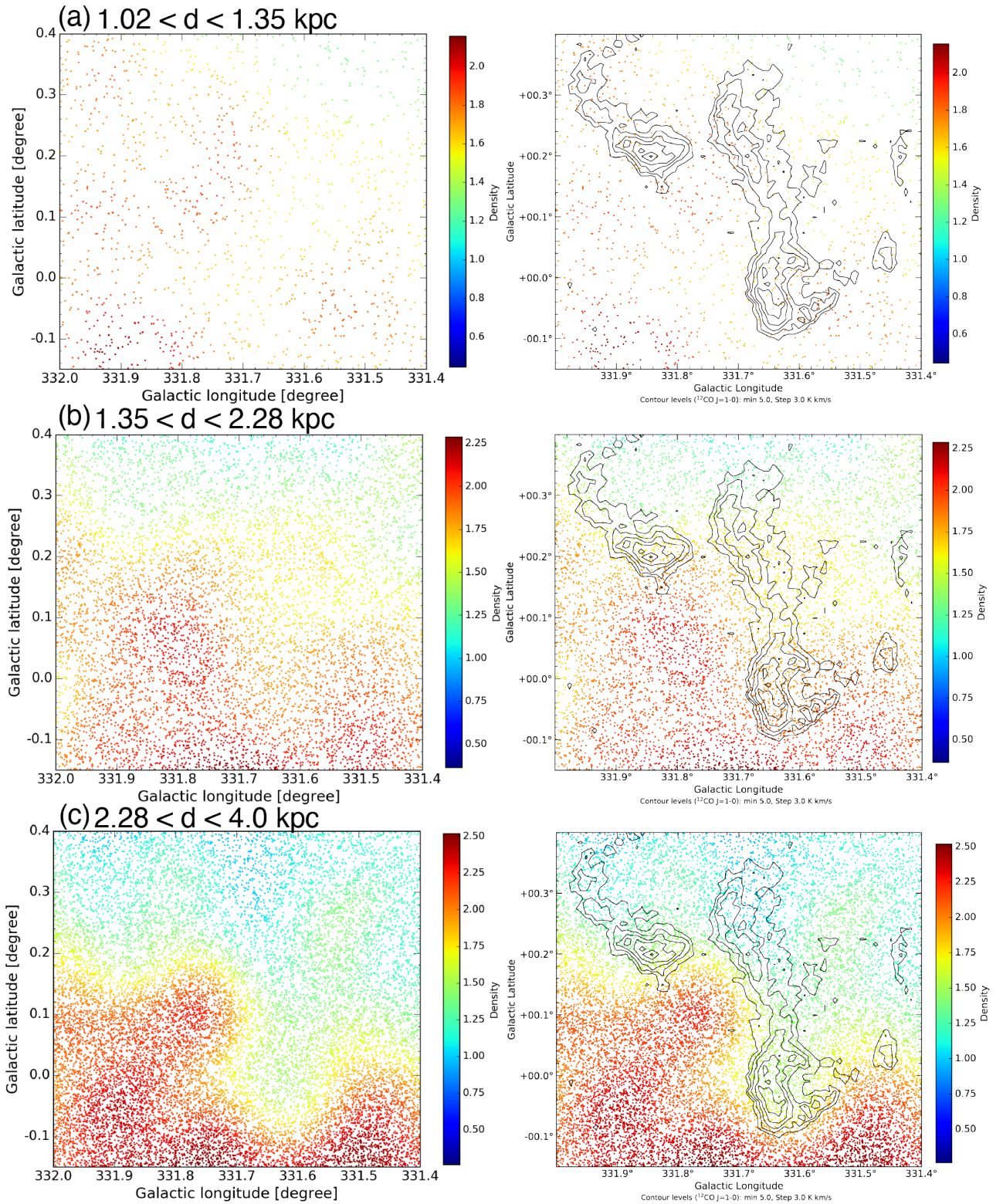


Fig. 8. The stellar density map of fixed stars obtained by Gaia DR3. The panels (a), (b), and (c) show from 1.02 kpc to 1.35 kpc, from 1.35 kpc pc to 2.28 kpc, and from 2.28 kpc pc to 4.0 kpc, respectively. The black contours in the right column are the same as Figure 1 (a). Alt text: The stellar density map of fixed stars obtained by Gaia DR3.

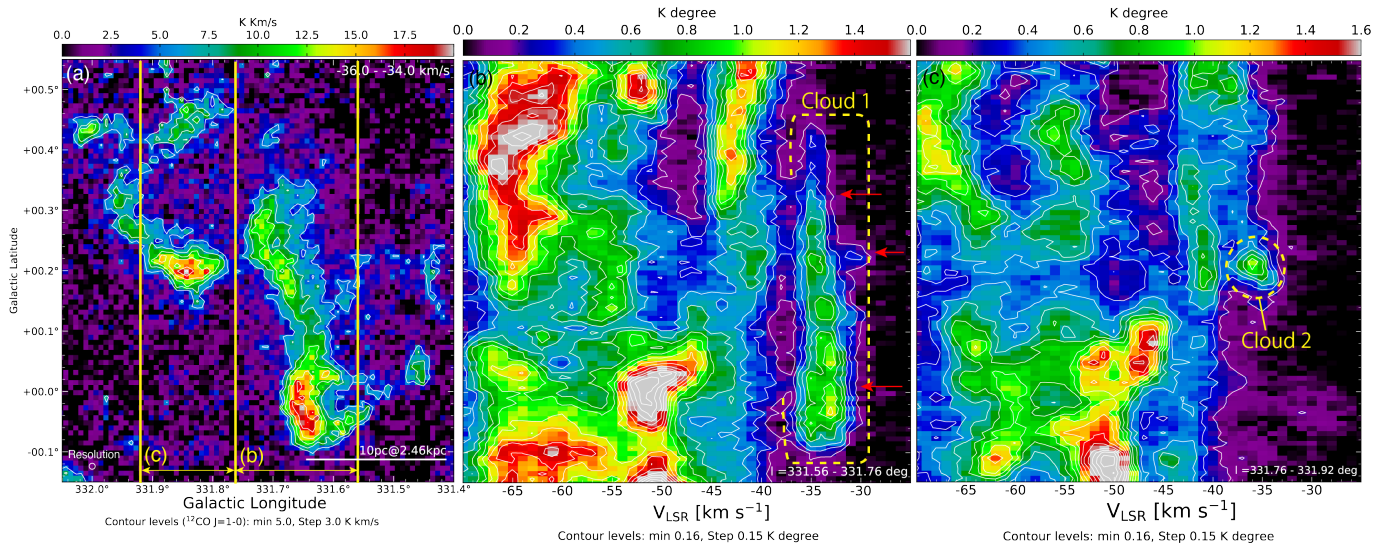


Fig. 9. (a) Same as Figure 1(a). The yellow lines show the integrated longitude ranges in the panels (b) and (c). (b) The $^{12}\text{CO } J = 1-0$ latitude-velocity diagram of Cloud 1. The red arrows indicate the bridge features connecting to the disk velocity component. The integrated longitude range is from $331.^{\circ}56$ to $331.^{\circ}76$. (c) The $^{12}\text{CO } J = 1-0$ latitude-velocity diagram of Cloud 2. The integrated longitude range is from $331.^{\circ}76$ to $331.^{\circ}92$. The lowest contour levels and intervals are 0.16 K degree and 0.15 K degree, respectively. Alt text: Mopra 12CO integrated intensity map, latitude-velocity diagram of Cloud 1, and latitude-velocity diagram of Cloud 2.

5 Conclusion

The conclusions of this paper are as follows:

1. Two head-tail molecular clouds were identified at $l = 331.^\circ 6$ and $b = 0.^\circ 0$. A detailed analysis of the CO data obtained by Mopra revealed that the two head-tail molecular clouds have the radial velocity of -35 km s^{-1} , which is different from the velocity components of the Scutum–Centaurus and Norma arms of the Milky Way in the line of sight at $l = 331.^\circ 6$.
2. The kinetic temperatures of the head-tail clouds were estimated from an analysis of the emission line intensity ratio of the APEX $^{13}\text{CO } J = 2-1$ and the Mopra $^{13}\text{CO } J = 1-0$ data. The heads and tail parts were $T_{\text{kin}} = 30-50 \text{ K}$, which is more heated than the quiescent molecular gas without heating sources. A comparison with the Herschel $160 \mu\text{m}$ data revealed no infrared source as heating sources corresponding to the entire head-tail molecular clouds.
3. We find a G-band extinction corresponding to the head-tail molecular cloud by analyzing the Gaia Data Release 3. Based on the breaking points analysis and morphological correspondence of interstellar extinction, the head-tail molecular clouds are likely to exist at a distance of $2.46 \pm 0.18 \text{ kpc}$ in the foreground of the RCW 106 GMC complex. The masses of Cloud 1 and Cloud 2 are derived to be $4.8 \times 10^3 M_\odot$ and $3.5 \times 10^3 M_\odot$, respectively.
4. We suggest that these head-tail moleclouds are falling clouds onto the Milky Way disk, and are heated by the shock compression with the interaction of the gas disk component. The shock velocity inferred is more than 35 km s^{-1} for the typical observed velocity of the IVCs, $40-60 \text{ km s}^{-1}$, which explains the high temperatures of the shocked gas in the present falling clouds. The present work revealed the falling molecular clouds for the first time based on direct observational signatures, which include the moving direction toward the plane by the head-tail morphology and high velocity similar to the IVCs.

Acknowledgements

We are grateful to the anonymous referee for carefully reading our manuscript and giving us thoughtful suggestions, which greatly improved this paper.

We utilized the Astropy (Astropy Collaboration et al. 2013; Astropy Collaboration et al. 2018; Astropy Collaboration et al. 2022), NumPy (van der Walt et al. 2011), Matplotlib (Hunter 2007), IPython (Perez, & Granger 2007), Miriad (Sault et al. 1995), and APLpy (Robitaille & Bressert 2012).

The authors are grateful to Dr. Michael Burton of the University of New South Wales and the Armagh Observatory and Planetarium for the archival CO survey data with Mopra. We also thank Dr. Graeme Wong for kindly supporting remote observations from Nagoya University.

This work was supported by Grants-in-Aid for Scientific Research (KAKENHI) from Japan Society for the Promotion of Science (JSPS), Grant Numbers JP15H05694, JP20H01945, JP21H00040, and JP22H00152. The Mopra radio telescope is part of the Australia Telescope National Facility, which is funded by the Australian Government for operation as a National Facility managed by CSIRO. The University of New South Wales Digital Filter Bank used for the observations with the Mopra Telescope was provided with support from the Australian Research Council.

This publication is based on data acquired with the Atacama Pathfinder Experiment (APEX) under programmes 092.F-9315

and 193.C-0584. APEX is a collaboration among the Max-Planck-Institut für Radioastronomie, the European Southern Observatory, and the Onsala Space Observatory. The processed data products are available from the SEDIGISM survey database located at <https://sedigism.mpifr-bonn.mpg.de/index.html>, which was constructed by James Urquhart and hosted by the Max Planck Institute for Radio Astronomy

The Herschel spacecraft was designed, built, tested, and launched under a contract to ESA managed by the Herschel/Planck Project team by an industrial consortium under the overall responsibility of the prime contractor Thales Alenia Space (Cannes), and including Astrium (Friedrichshafen) responsible for the payload module and for system testing at spacecraft level, Thales Alenia Space (Turin) responsible for the service module, and Astrium (Toulouse) responsible for the telescope, with in excess of a hundred subcontractors.

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